



Hailemichael Atrsaw Tibebu

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ABOUT ME

DevOps Engineer with years of experience specializing in multi-cloud infrastructure automation and **Kubernetes** orchestration. Expert in building custom **Python-based** tooling and architecting **self-hosted RKE2 clusters**. Proven track record in streamlining **CI/CD pipelines** and automating complex database lifecycles across multiple cloud providers.

WORK EXPERIENCE

JUNIOR SOFTWARE DEVELOPER – TEFER (CONTRACTED BY STABELDUEL BASED IN US) – 29/04/2020 – 10/06/2021 – ADDIS ABABA, ETHIOPIA

Full-Stack Development: Developed web applications using the Django framework and managed relational databases with PostgreSQL.

Feature Implementation: Contributed to the development of core business modules, including user authentication, authorization, and automated data reporting.

Bug Resolution & Maintenance: Identified and resolved technical issues in legacy codebases, improving overall system stability and performance.

Collaborative Development: Actively participated in code reviews and agile ceremonies, contributing to the improvement of internal coding standards and best practices.

BACKEND DEVELOPER – TEFER (CONTRACTED BY STABELDUEL BASED IN US) – 11/06/2021 – 03/08/2023 – ADDIS ABABA, ETHIOPIA

API Development: Built robust backend systems using Django Rest Framework and PostgreSQL, focusing on scalability and performance.

Asynchronous Processing: Implemented background task processing using Celery and Redis to handle long-running operations and improve user experience.

Containerization: Streamlined development and deployment workflows by containerizing applications using Docker and Docker Compose.

Database Optimization: Optimized complex SQL queries and database schema designs, significantly improving API response times and system throughput.

Code Quality & Testing: Implemented comprehensive unit and integration tests using Pytest, ensuring high code reliability and reducing production regressions.

BACKEND | CLOUD DEVELOPER – TEFER (CONTRACTED BY STABELDUEL BASED IN US) – 04/08/2023 – 07/07/2024 – ADDIS ABABA, ETHIOPIA

Business or Sector: Information and communication | **Department:** Software Development | **Website:** <https://www.tefer.io/>

AWS Infrastructure & Scalability: Designed, provisioned, and scaled AWS infrastructure (EC2, S3, RDS, VPC, ELB, Lambda, SNS, SQS) using IaC principles; optimized application delivery with Nginx and Auto Scaling Groups.

Scalable API Development: Built robust REST APIs using Django Rest Framework and managed PostgreSQL databases, ensuring data integrity and high availability.

Automation & Performance: Automated complex workflows and background tasks with Python and Celery; implemented Redis caching and messaging to enhance system performance.

Containerization & CI/CD: Containerized and orchestrated applications using Docker and Docker Compose; managed CI/CD pipelines and code repositories via Azure DevOps Services.

Data Lifecycle & Reliability: Automated RDS backup schedules and implemented S3 lifecycle policies to optimize storage costs; maintained production environments ensuring 99.9% uptime.

DEVOPS ENGINEER – EXPONENT.CH – 08/07/2024 – Current – GENEVA, SWITZERLAND

Business or Sector: Information and communication | **Department:** Infrastructure Management |
Website: <https://exponent.ch/en>

Multi-Cloud Infrastructure Automation: Orchestrated and managed complex infrastructure across Infomaniak, Exoscale, and managed Kubernetes services (EKS, AKS, GKE) using Terraform, Terragrunt, and Pulumi.

Accelerated Deployment Cycles: Reduced application deployment time to under 2 hours by developing a reusable, generic Helm chart and building secure CI/CD pipelines with GitLab CI and GitHub Actions.

Enhanced Observability: Architected and deployed a comprehensive monitoring stack (Prometheus, Grafana, Loki, Mimir) for self-hosted RKE2 clusters, providing deep visibility into system metrics and logs.

Secrets & Security Management: Centralized secrets management using Infisical and 1Password; implemented Renovate bot to automate dependency updates and security patching across infrastructure repositories.

Traffic & Load Balancing: Implemented robust load balancing and service mesh solutions using Envoy to ensure high availability and efficient traffic routing.

Custom Tooling & Automation: Developed a "Sentry Gateway" in Python (Litestar) to aggregate and transform Prometheus alerts into actionable Sentry issues, improving developer response times.

Open Source Engagement: Contributed to open-source projects, focusing on improving DevOps tooling and infrastructure automation workflows.

● **EDUCATION AND TRAINING**

08/05/2023 – CURRENT Addis Ababa
AWS CLOUD COMPUTING ALX

Website <https://www.alxafrica.com/> | **Field of study** Software and applications development and analysis

01/10/2016 – 23/02/2021 Addis Ababa, Ethiopia
ELECTRICAL AND COMPUTER ENGINEERING (COMPUTER ENGINEERING FOCUS AREA) Jimma University

Website <https://ju.edu.et/> | **Field of study** Engineering and engineering trades not further defined | **Final grade** 3.60 |

Level in EQF EQF level 4 | **Type of credits** ECTS

● **SKILLS**

Cloud Service Providers

AWS | Azure | GCP | Exoscale | DigitalOcean | Cloudflare | Infomaniak

Programming Languages

Python

DevOps

Docker | Kubernetes | Terraform | Helm | Gitlab CI | GitHub Actions | Terragrunt | Ansible | Teleport | Infisical | Envoy

- Database
 - PostgreSQL | MySQL
- Backend Frameworks
 - Django REST Framework | Flask
- Monitoring and Logging
 - Grafana | Prometheus | Loki | Mimir

● **HONOURS AND AWARDS**

04/02/2023
AWS Certified Solutions Architect - Associate – Amazon Web Services

Link <https://www.credly.com/badges/4898bdae-19da-4ce7-b501-4ef43db0737e>

18/10/2023
AWS Certified Cloud Practitioner – Amazon Web Services

Link <https://www.credly.com/badges/93a90952-c945-4aa1-898f-72a957e58ea9>

09/11/2023
IELTS – British Council

Overall band score: 7.0

21/05/2024
Verified International Academic Qualifications – World Education Services

Link https://www.credly.com/badges/bd73fccc-196b-469a-bbb3-9c117ede0373/linked_in_profile

● **LANGUAGE SKILLS**

Mother tongue(s): **AMHARIC**
Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	B2	C1	C1	B2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

● **PROJECTS**

08/07/2024 – CURRENT
XKS (Kubernetes Provisioning Engine)

Developed a sophisticated **Python-based** provisioning engine designed to automate the deployment of **high-availability RKE2 clusters** across multiple cloud providers, including **Infomaniak** and **Exoscale**. The tool orchestrates the entire infrastructure lifecycle, from creating **Envoy-based load balancers** and configuring agent/server nodes to generating secure kubeconfig certificates. It further streamlines cluster readiness by integrating essential post-deployment stacks such as Cloud Controllers, Cilium CNI, and CoreDNS, while ensuring operational stability through custom watcher pods for health monitoring and automated port management for LoadBalancer services.

08/07/2024 – CURRENT
Datacycle (Database Automation Platform)

Engineered Datacycle, a sophisticated database automation platform developed with **Python (Litestar and Mesop)** to streamline the management of **CloudNativePG (CNPG) clusters**. The platform implements an auto-discovery mechanism to identify and monitor configured CNPG instances, providing a centralized interface to visualize database and backup history, including retention metrics and timestamps. It offers flexible backup orchestration, allowing for manual triggers of single or bulk database backups alongside a cron-scheduled worker for automated, encrypted cloud

uploads. Additionally, Datacycle enables seamless restoration of backups to new databases across instances and features a proactive validation worker that verifies backup integrity through automated trial restores, ensuring high availability and data reliability.

08/07/2024 – CURRENT

Chart-Generic (Standardized Helm Chart)

Created a highly configurable and standardized **Helm** chart to unify application deployments across **Kubernetes** environments, significantly reducing the overhead of onboarding new services. This generic chart provides a robust template supporting complex configurations for **Deployments, Horizontal Pod Autoscalers (HPA), Network Policies, Ingress, and Jobs**, ensuring consistent deployment patterns and best practices across the organization.